Distributed Database Systems IT 504 (3-0-2-4)

MTech ICT-SS Core Elective

Prerequisites: Fundamental Course on DBMS

Instructor: Minal Bhise **minal_bhise@dau.ac.in**

Objectives:

Collection of correlated databases distributed over a cluster of servers can facilitate scalable query processing. Course includes foundational work, recent developments and trends in scalable database management systems. Modern Data Storage and Query processing for Parallel and distributed databases will be discussed. Students will be working on research projects in the domain of modern distributed data storage and query processing during labs. Students will be using various SQL and NoSQL database management tools for implementing their research ideas.

Resources:

- Tamer Oszu and Patrick Valduriez, Principles of Distributed Systems, Fourth Edition, Springer 2020
- Silberschatz, Korth & Sudarshan, Database System Concepts, Seventh Edition, 2019,McGraw-Hill
- Research Papers

Course Outcomes:

The course will make students understand various fundamental concepts on which Distributed Databases work. The students will have some hands-on experience with distributed database technology and tools through a series of assignments and a project. They will also learn some NoSQL technologies like Graph tools.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X			X	X	X		X

Course Plan:

Topics	Lecture hrs
Overview, Basic definitions	2
Relational Model basics: Relational Algebra, SQL, Query Processing	4
Introduction of Distributed Databases, Distributed architecture, Distributed Storage	4
distributed database design considerations and strategies: fragmentation, distribution and replication models	8
Distributed and Parallel Query Processing, scalability	2
Query Localization, Query Optimization query cost estimation	6
distributed transaction management: concurrency control, crash recovery	4
various data models: SQL and NoSQL databases, RDF and graph databases	4
research issues in modern distributed database management	6

Evaluation Scheme:

- InSem Exam: 25%
- Lab Assignments: 35%
- Final Exam: 40%

The Lab Assignments will be evaluated in the form of lab vivas, presentations, demonstrations, and write-ups spread all over the semester.